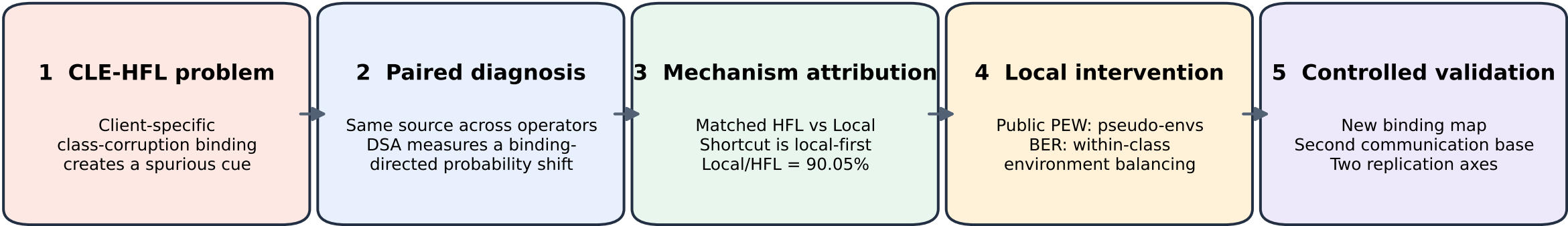
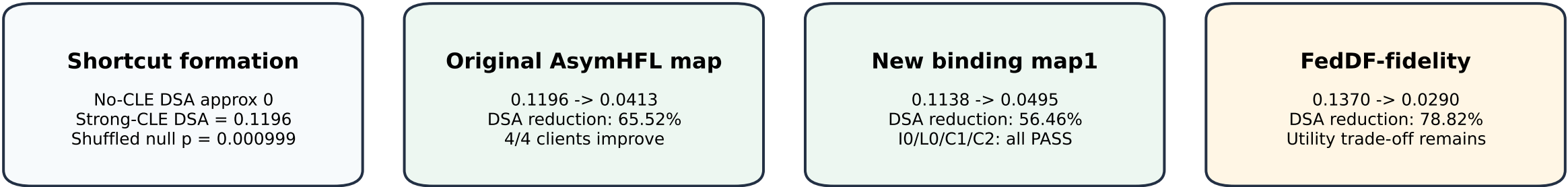


# CLE-HFL: from directional shortcut identification to controlled mitigation

A problem-diagnosis-attribution-intervention-validation evidence chain



## Formal evidence



## Claim boundary

Supported: directional shortcut formation, local-first attribution, and mitigation replicated across binding maps and communication bases.  
Not supported: taxonomy-free discovery, universally lossless utility, or generalization across partitions, seeds, datasets, and real domains.